U O W

CSIT305 Emerging Information Technologies and their Applications

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IoT and Edge Computing

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Subject Logistics

(1) Advanced Database Overview
(2) Spatial Data and Computing
(3) Graph Data and Computing
(4) Text Data and Processing

Multi-Modal Database

(1) AI4SE: artificial intelligence for software engineering
(2) Software Engineering in Cloud Computing I
(3) Software Engineering in Cloud Computing II

Software

(1) IoT and Edge Computing
(2) Digital Twin Technology
(3) Blockchains and Smart Contracts
(4) Emerging Cyber-Attacks

Networking

Outline

- Internet of Things
- Edge Computing
- Applications

Internet of Things

- The Internet of things (IoT) is the inter-networking of physical devices, vehicles (also referred to as "connected devices" and "smart devices"), buildings, and other items embedded with electronics, software, sensors, actuators, and network connectivity which enable these objects to collect and exchange data.

- The internet of things, or IoT, is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

- Internet of Things (IoT) is a system of devices connected to the Internet with the ability to collect and exchange data from users or environment with no human intervention.

Things

- Physical objects that are connected to the internet and can collect, exchange, or act on data.
  - Smart home devices: smart speakers, thermostats, lights, locks, cameras.
  - Wearables: fitness trackers, smartwatches, health monitors.
  - Industrial sensors: temperature, pressure, vibration, chemical detectors.
  - Vehicles: connected cars, fleet trackers, autonomous drones.
  - Healthcare devices: remote patient monitors, insulin pumps, smart inhalers.
  - Agriculture sensors: soil moisture sensors, smart irrigation systems.
  - Smart city infrastructure: streetlights, parking meters, traffic sensors.

IoT Systems

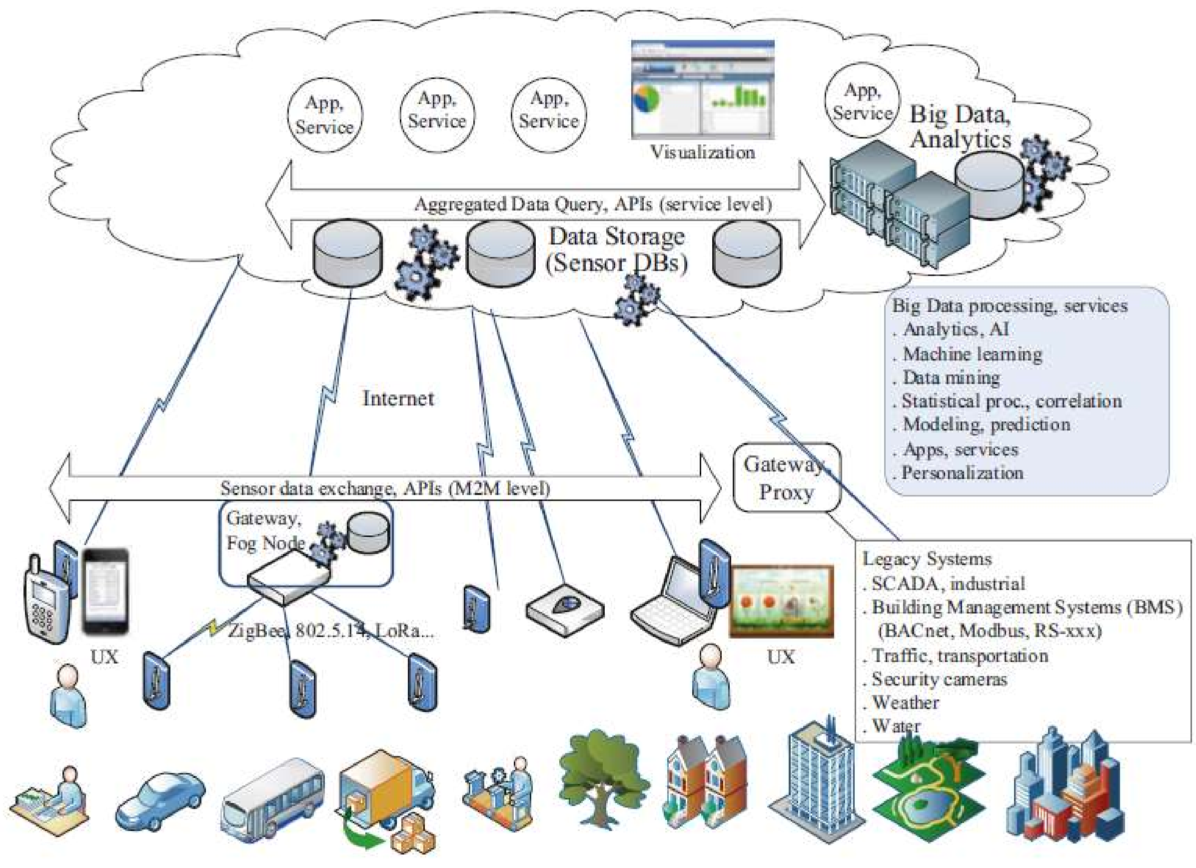

Fig. 1.1 IoT systems

Cloud

Communications layer

Edge components

IoT Ecosystem

- Sensors
  - Embedded systems, real-time operating systems, energy-harvesting sources, Micro-Electro-Mechanical Systems (MEMs).
- Sensor communication systems
  - Wireless personal area networks reach from 0 cm to 100 m. Low-speed and low-power communication channels, often non-IP based have a place in sensor communication.
- Local area networks
  - Typically, IP-based communication systems such as 802.11 Wi-Fi used for fast radio communication, often in peer-to-peer or star topologies.
- Aggregators, routers, gateways
  - Embedded systems providers, cheapest vendors (processors, DRAM, and storage), module vendors, passive component manufacturers, cellular and wireless radio manufacturers, edge analytics packages, edge security providers, certificate management systems.
- WAN
  - Cellular network providers, satellite network providers, Low-Power Wide-Area Network (LPWAN) providers. Typically using internet transport protocols targeted for IoT and constrained devices like MQTT, CoAP, and even HTTP.

IoT Ecosystem

- Cloud
  - Infrastructure as a service provider, platform as a service provider, database manufacturers, streaming and batch processing manufacturers, data analytics packages, software as a service provider, data lake providers, Software-Defined Networking/Software-Defined Perimeter providers, and machine learning services.
- Data analytics
  - As the information propagates to the cloud en-mass. Dealing with volumes data and extracting value is the job of complex event processing, data analytics, and machine learning techniques.
- Security
  - Tying the entire architecture together is security. Security will touch every component from physical sensors to the CPU and digital hardware, to the radio communication systems, to the communication protocols themselves. Each level needs to ensure security, authenticity, and integrity. There cannot be the weak link in a chain, as the IoT will form the largest attack surface on earth.
- Applications
  - Where end users or organisations interact with IoT systems.

Why IoT?

- Data-driven insights
- Automation and efficiency
- Cost reduction
- Improved safety and quality of life
- Scalability and real-time action
- New business models

Why Now?

- Industry 4.0 and digitalization
- Sensors – installed base, variety, lower cost, and easier integration
- Smart phones – sensors, gateways, and UI devices
- Cloud computing – global and capacity on demand
- AI and ML technologies – actionable insights with IoT data
- Internet – technology, global infrastructure, and users

IoT Example - The Refrigerator

- A refrigerator is just a thing, so it can be anything besides a computer.
- If it's a computer, we're not calling that a thing, but anything besides a computer.
- Now the next thing you do is you add to that some type of computational intelligence (Micro controller).
- So that's intelligent refrigerator. But maybe you wouldn't call that Internet of Things, because it's not actually networked yet. Then, to top it off, you add Internet connectivity.
- It's got some kind of computation inside and it's got a network connection which means it can use all sorts of other resources that are not local.

IoT Example - What the IoT Refrigerator can do

- You can have IoT fridge detect that you are low on butter and then just order that, and then the next morning, butter will appear on your doorstep.
- You can search for low food prices.
- Give you a good idea for food for you tomorrow for the meal.
- IoT device should have a simple interface, have computational intelligence and be connected to the internet.

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How to build an IoT system?

- Step 1: Define the Purpose and Requirements
- Before touching any hardware or code:
  - Identify the problem -> What will your IoT system monitor or control? (e.g., smart agriculture, health monitoring, industrial sensors)
  - Set functional goals -> Data to collect, response time, accuracy, security.
  - Decide constraints -> Budget, power consumption, network availability, device size.

How to build an IoT system?

- Step 2: Choose the Hardware Layer (Things)
- This layer includes sensors, actuators, and edge devices.
- Key elements:
  - Sensors -> Measure temperature, humidity, motion, GPS, etc.
  - Actuators -> Perform actions (motors, relays, lights).
  - Microcontroller or SBC (Single-Board Computer):
    - Low-power: Arduino, ESP32, STM32.
    - More processing: Raspberry Pi, Jetson Nano.
  - Connectivity modules -> Wi-Fi, LoRa, Zigbee, Bluetooth, 4G/5G, NB-IoT.

How to build an IoT system?

- Step 3: Implement Connectivity & Communication
- IoT devices need to send data reliably and efficiently.
- Options:
  - Protocols (Device <-> Cloud):
    - MQTT -> Lightweight, good for low-bandwidth.
    - HTTP/HTTPS -> Common, but heavier.
    - CoAP -> Constrained devices.
  - Network Types:
    - Short-range: Wi-Fi, Bluetooth, Zigbee.
    - Long-range: LoRaWAN, NB-IoT, 4G/5G.

How to build an IoT system?

- Step 4: Data Processing Layer (Edge or Cloud)
- You can process data locally (edge computing) or in the cloud.
- Edge Computing:
  - Immediate response.
  - Less cloud dependency.
  - Examples: ESP32 doing local analytics, Raspberry Pi running AI models.
- Cloud Processing:
  - Centralized storage and analytics.
  - Scalable.
  - Platforms: AWS IoT Core, Azure IoT Hub, Google Cloud IoT, ThingsBoard.

How to build an IoT system?

- Step 5: Software & Integration
  - Firmware -> Runs on IoT devices to collect data and send it.
  - IoT Middleware -> Manages communication, authentication, and data routing.
  - APIs -> Connect IoT system with apps, dashboards, and third-party tools.
  - Data Storage -> SQL (structured) or NoSQL (time-series, unstructured).
  - Visualization -> Grafana, Power BI, custom dashboards.

How to build an IoT system?

- Step 6: Security
- IoT systems are often the weakest cybersecurity link.
  - Use TLS/SSL encryption for data in transit.
  - Implement device authentication (certificates, tokens).
  - Keep firmware updated.
  - Limit open network ports and unnecessary services.

How to build an IoT system?

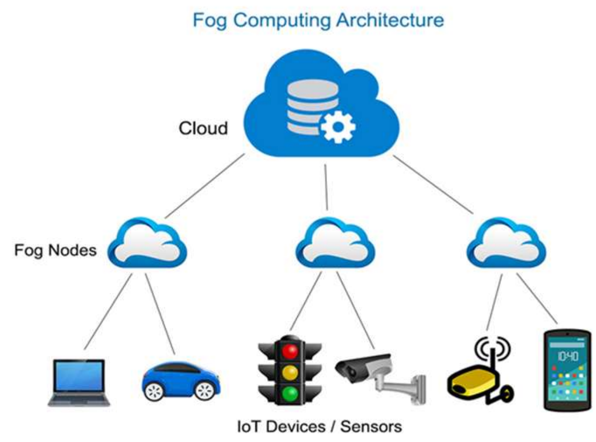
- https://www.youtube.com/watch?v=7QzrjSPBgBo

Edge Computing

- Edge computing is a decentralized computing paradigm that involves bringing data processing and storage closer to the point of data generation.
- In traditional cloud computing models, data is transmitted to remote servers or data centers for processing and analysis. In contrast, edge computing conducts processing at or near the edge devices, such as sensors, IoT devices, or edge servers, located in proximity to the data source.
- Edge computing is to diminish latency and bandwidth requirements by locally processing data at the edge rather than shuttling it back and forth to a centralized cloud infrastructure.
- By handling data closer to its origin, edge computing enables real-time or near-real-time analysis and response, proving particularly advantageous for time-sensitive applications or situations with limited or unreliable network connectivity.

What is Edge Computing?

- Fog computing extends the concept of edge computing
- The term "Fog Computing" was introduced by the Cisco Systems
- Fog nodes take place on the responsibilities of processing and analysis
  - Edge Router with intelligence (e.g., Cisco IOS router) to decide what data needs to be sent to a data center (complex functions), and what data needs to be processed locally (simpler tasks)



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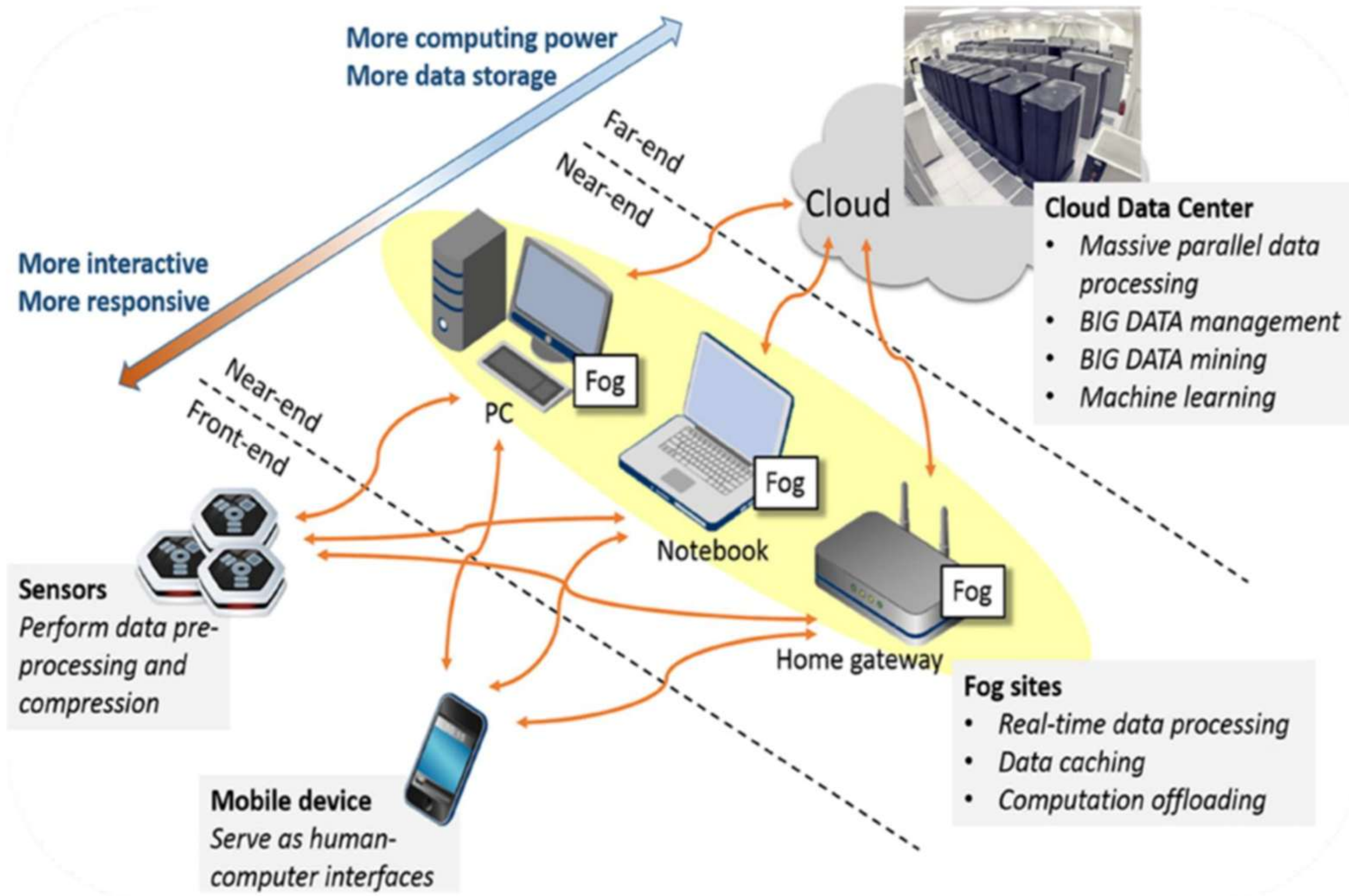
Edge Devices

- Edge devices are the actual physical hardware at the “edge” of the network that perform data collection, local processing, and sometimes decision-making before sending data to the cloud.
  - Sensors & Actuators
  - IoT gateways
  - Edge servers / mini data centers
  - Smart end devices (with on-board intelligence)

Example



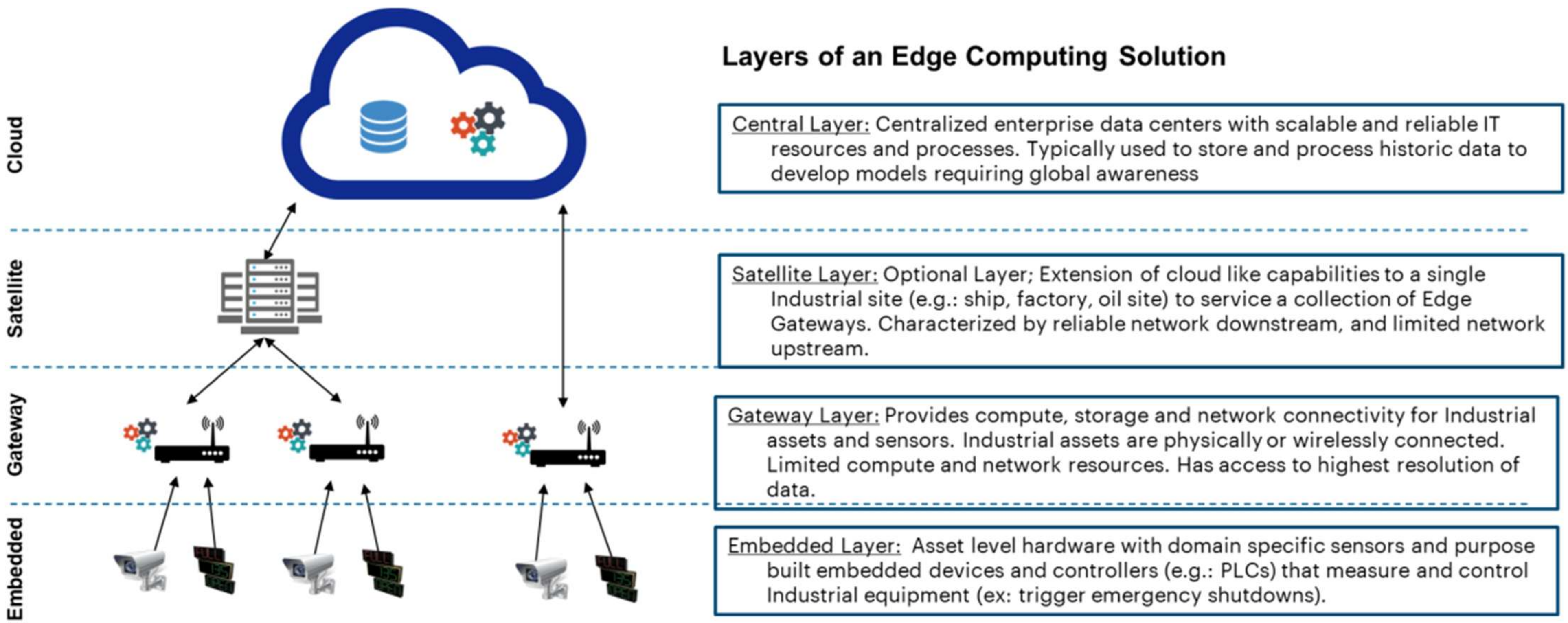
More computing power
More data storage
More interactive
More responsive
Far-end
Near-end
Near-end
Front-end
Cloud
Cloud Data Center
• Massive parallel data processing
• BIG DATA management
• BIG DATA mining
• Machine learning
PC
Fog
Notebook
Fog
Home gateway
Fog
Fog sites
• Real-time data processing
• Data caching
• Computation offloading
Sensors
Perform data pre-processing and compression
Mobile device
Serve as human-computer interfaces

Why Requires Edge Computing?

- Edge computing was developed to address applications and services that do not fit the paradigm of the cloud
- Edge Computing keeps data right where the Internet of Things needs it
- Existing data protection mechanisms in Cloud Computing are insufficient

Edge Computing

- Edge computing extends control to the source of internet of things.



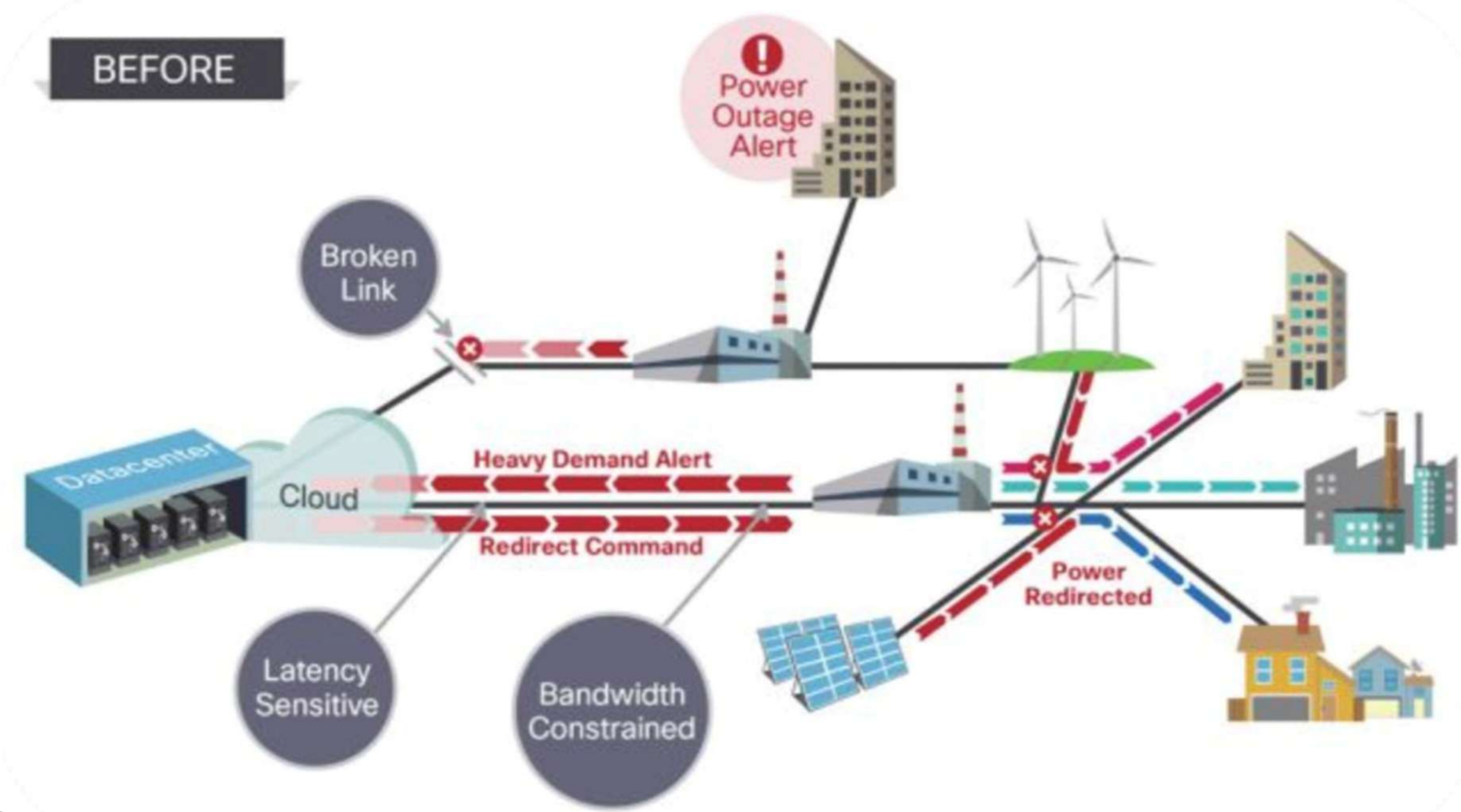
Edge Computing

- The advantages offered by Edge computing are as follows:
  - Reduced Latency
    - Through local data processing, edge computing minimizes the round-trip time for data between edge devices and the cloud. This is important for applications demanding immediate or nearly immediate processing, including scenarios like autonomous vehicles, industrial automation, or real-time monitoring systems.
  - Bandwidth Optimization
    - Edge computing optimizes bandwidth utilization and diminishes data transmission costs by decreasing the volume of data transferred to the cloud. This holds particular importance in situations where network connectivity is restricted or comes at a high cost.

Edge Computing

- The advantages offered by Edge computing are as follows:
  - Enhanced Privacy and Security
    - Edge computing facilitates local data processing, minimizing the necessity of transmitting sensitive data to the cloud. This fortifies privacy and security by maintaining data in close proximity to its source, thereby reducing the vulnerability to potential security breaches or data leaks.
  - Offline Operation
    - Edge computing empowers applications to function seamlessly even in the absence of network connectivity. Local processing and storage capabilities enable devices to operate autonomously, guaranteeing uninterrupted functionality in remote or disconnected environments.

Edge Example - Before



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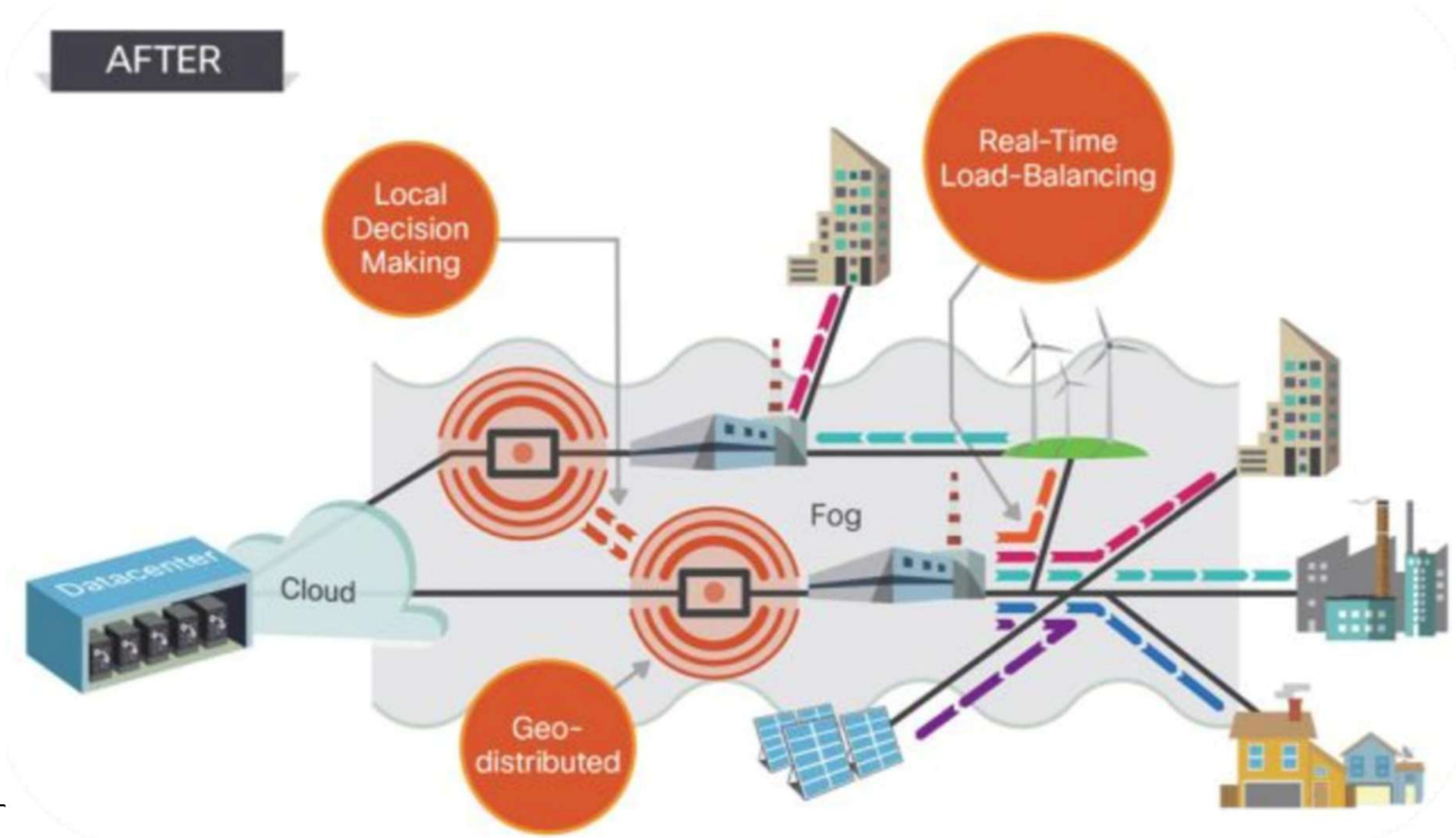
Edge Example - After



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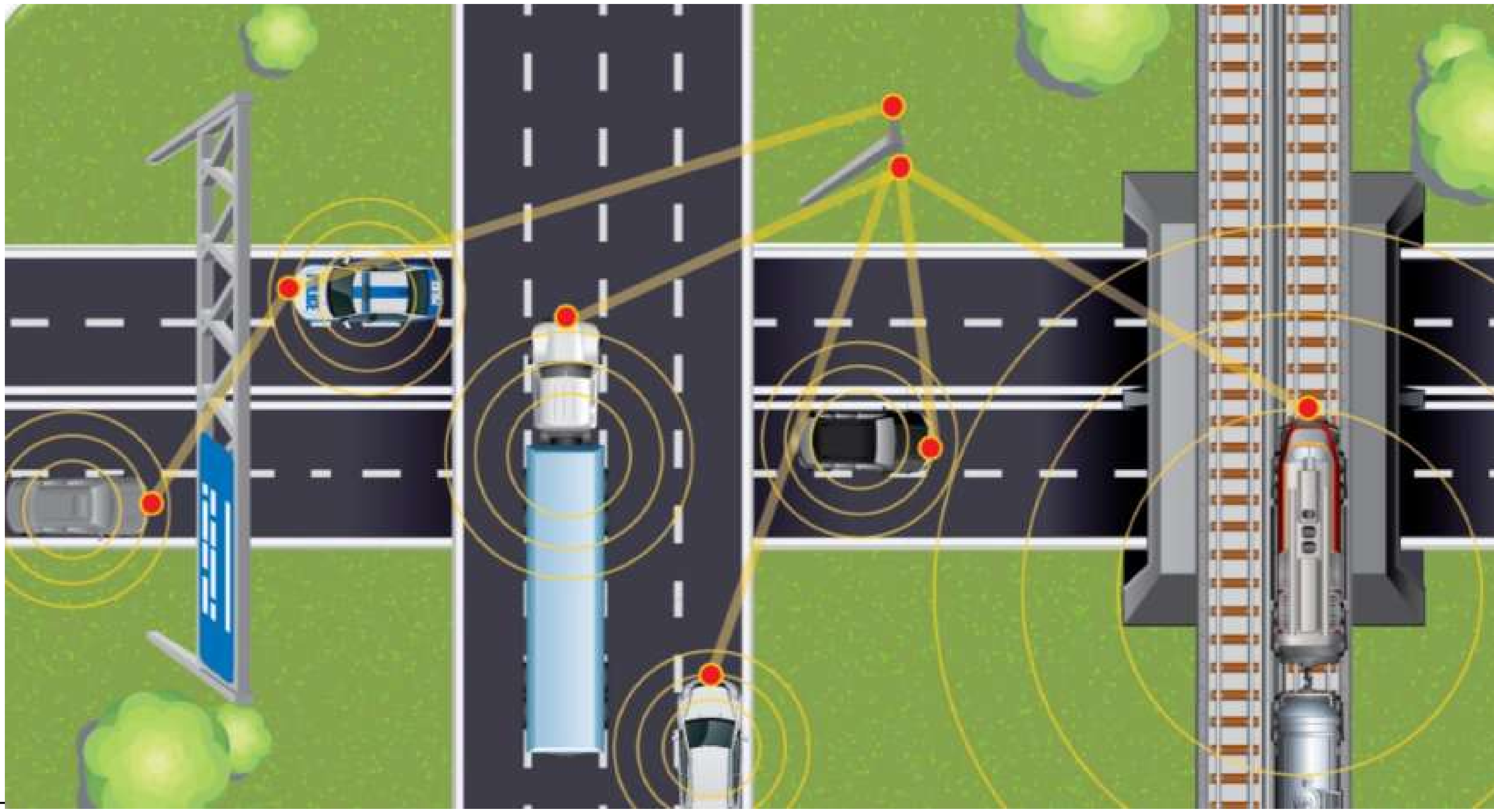
Another Example



Cloud Computing vs. Edge Computing vs. Fog Computing

- Cloud computing refers to the centralized storage and processing of data on remote servers or data centers. In the cloud model, data is sent over the network to the cloud infrastructure for processing, analysis, and storage. Cloud computing provides virtually unlimited storage capacity, high computational power, and scalability. It is well-suited for applications that require massive data storage, complex computations, and data processing that does not require real-time or low-latency response.

Cloud Computing vs. Edge Computing vs. Fog Computing

- Cloud computing
  - Advantages:
    - Scalability: Cloud computing provides elastic resources, allowing users to scale up or down based on their needs.
    - Accessibility: Data and applications can be accessed from anywhere with an internet connection.
    - Cost-effective: Users can avoid upfront hardware and infrastructure costs.
  - Limitations:
    - Latency: The round-trip time for data transmission between the edge devices and the cloud can introduce latency, which may not be suitable for real-time or latency-sensitive applications.
    - Bandwidth Requirements: Cloud computing relies on transferring large amounts of data over the network, which can be costly and inefficient in scenarios with limited bandwidth or high data volumes.

Cloud Computing vs. Edge Computing vs. Fog Computing

- Edge computing involves processing data at or near the edge devices or sensors, closer to the data source. By performing computations locally, edge computing reduces latency and minimizes the need for transferring data to the cloud. Edge computing is ideal for applications that require real-time or near-real-time processing, low latency, and efficient bandwidth usage.

Cloud Computing vs. Edge Computing vs. Fog Computing

- Edge computing
  - Advantages
    - Low Latency: Edge computing enables real-time or near-real-time processing, reducing the latency associated with cloud-based processing.
    - Bandwidth Optimization: Through local data processing, edge computing minimizes the amount of data that must be sent to the cloud, optimizing the utilization of bandwidth.
    - Offline Operation: Edge devices can continue to function even in scenarios with limited or no network connectivity.
  - Limitations
    - Limited Computational Power: Edge devices usually possess constrained processing power and storage capacity in comparison to cloud infrastructure, potentially limiting the complexity of computations that can be executed.
    - Scalability: Expanding edge computing infrastructure across a multitude of devices can pose challenges owing to complexities in management and coordination.

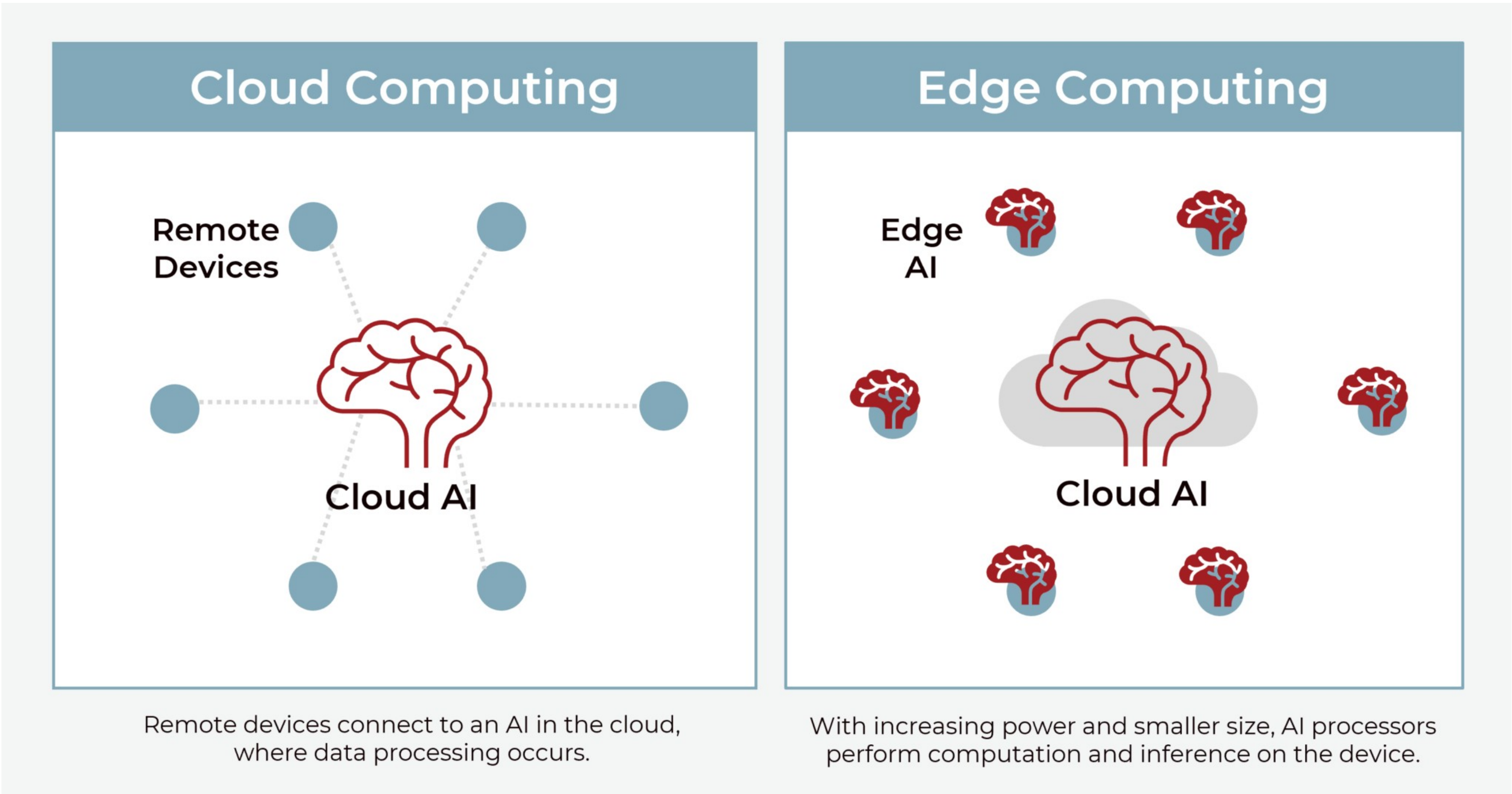
Cloud Computing vs. Edge Computing vs. Fog Computing

- Fog computing extends the concept of edge computing by introducing a hierarchical architecture that incorporates intermediate computing nodes, referred to as fog nodes, situated between edge devices and the cloud. These fog nodes take on the responsibilities of processing and analysis, bringing computational capabilities closer to the edge devices while maintaining the utilization of the cloud for specific tasks. Fog computing is well-suited for applications that demand a balance between low latency, local processing, and access to cloud resources.

Cloud Computing vs. Edge Computing vs. Fog Computing

- Fog computing
  - Advantages
    - Low Latency: Fog computing reduces latency by bringing processing closer to edge devices, similar to edge computing.
    - Scalability: Fog nodes can be deployed at various levels of the network hierarchy, providing scalability and management advantages compared to edge-only deployments.
    - Resource Offloading: Fog nodes can offload certain processing tasks to the cloud, using its higher computational power and storage capacity.
  - Limitations
    - Complexity: The hierarchical nature of fog computing introduces additional complexities in terms of management, coordination, and deployment
    - Network Congestion: The increased number of intermediate fog nodes can introduce network congestion and increase the complexity of data routing and communication.

When Edge Computing meets with AI



IoT in Manufacturing (IIoT)

- Improved visibility across your manufacturing operations—make more informed decisions with a real-time picture of operational status.
- Improved utilization—maximize asset performance and uptime with the visibility required for central monitoring and management.
- Reduced waste—take faster action to reduce or prevent certain forms of waste.
- Improved quality—detect and prevent quality problems by finding and addressing equipment issues sooner.
- https://www.youtube.com/watch?v=5W8MJuRx6Ck

IoT in Consumer

- Smart home gadgetry
  - Smart irrigation, smart garage doors, smart locks, smart lights, smart thermostats, and smart security.
- Wearables
  - Health and movement trackers, smart clothing/wearables.
- Pets
  - Pet location systems, smart dog doors.
- https://www.youtube.com/watch?v=sYqjs8TKkOE

IoT in Smart City

- Enhanced citizen safety
- Optimized public services
- Economic growth
- Improved traffic flow
- Improved environment
- https://www.youtube.com/watch?v=p88S5r22xOY

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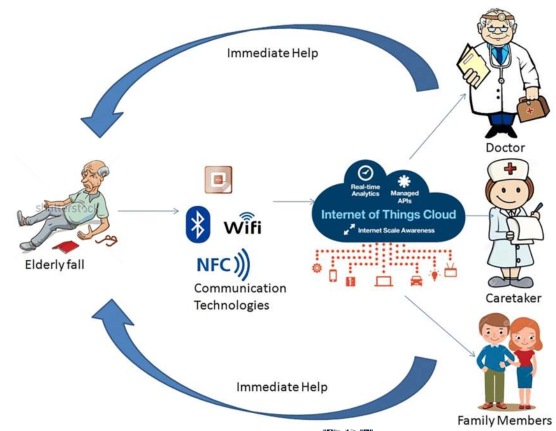
IoT in Retail

- Automated checkout systems
- Energy management
- Product tracking
- Environmental monitoring
- Smart shelves
- Personalised marketing
- https://www.youtube.com/watch?v=YTGHfFBPwgk

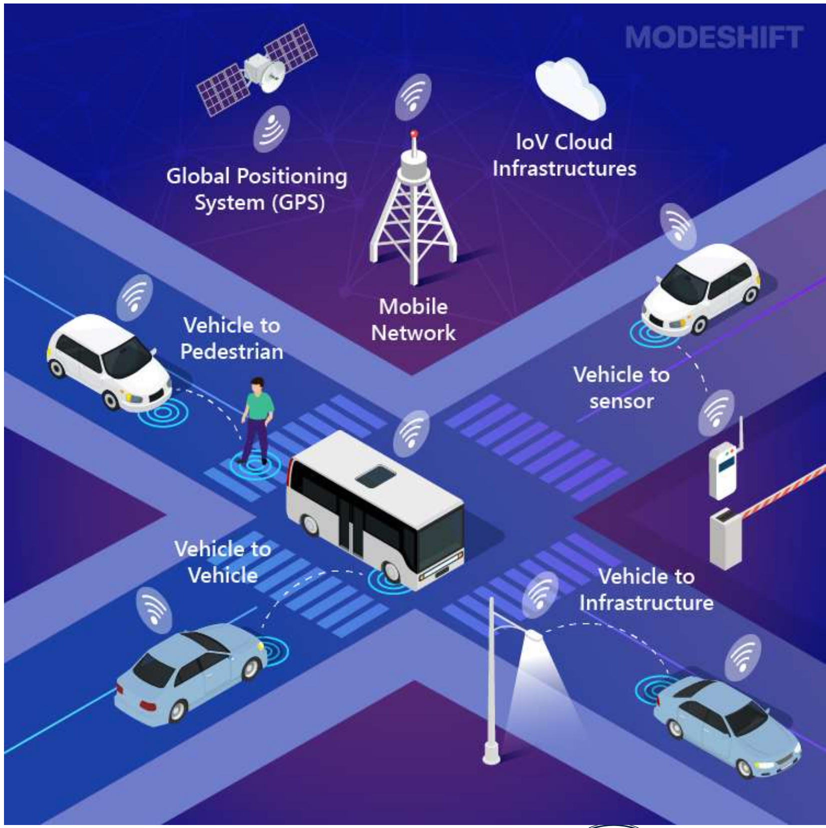
IoT in Healthcare

- In-home patient care
- Hospital equipment and supply asset tracking
- Pharmaceutical tracking and security
- IoT for physicians
- Drug research
- Patient fall indicators
- https://www.youtube.com/watch?v=PMhMOUp0knM



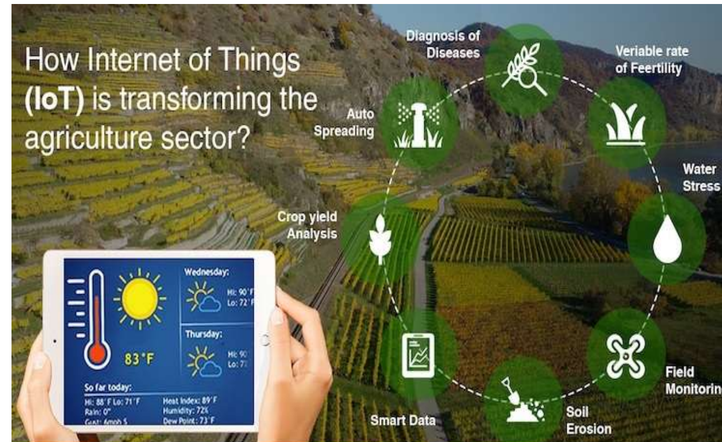
IoT in Intelligent Transportation Systems

- Autonomous vehicles
- Fleet tracking and location awareness.
- Asset and package tracking within fleets.
- Preventative maintenance of vehicles on the road.
- https://www.youtube.com/watch?v=V9vq45wnSoY



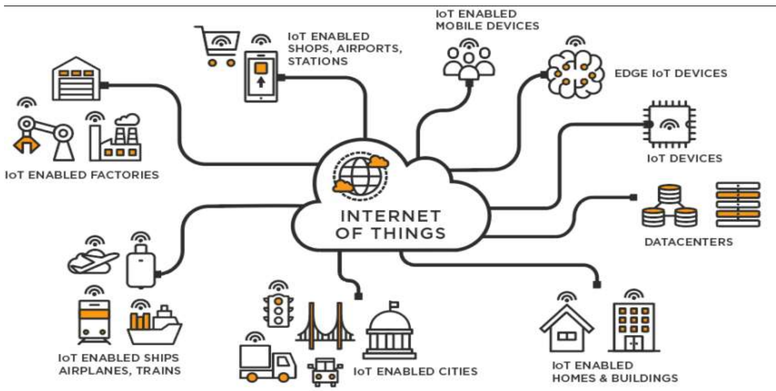
IoT in Agriculture and Environment

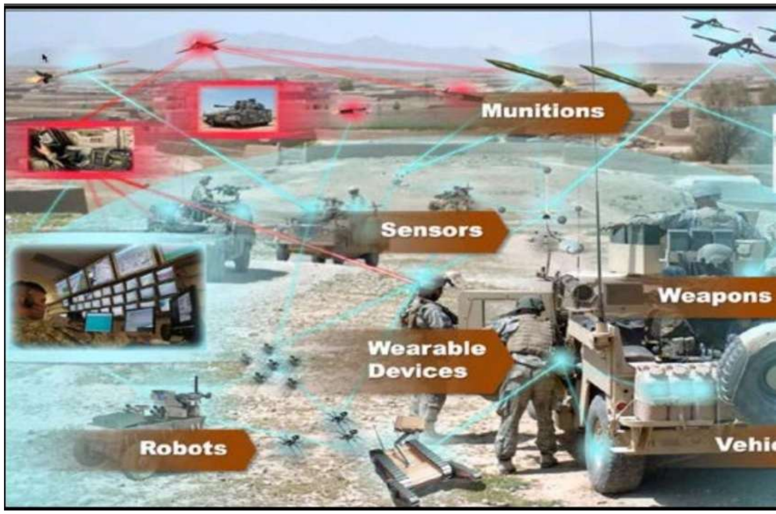
- Smart irrigation and fertilization techniques to improve yield.
- Smart lighting in nesting or poultry farming to improve yield.
- Livestock health and asset tracking.
- Drones-based land surveys.
- Robotic farming.
- Volcanic and fault line monitoring for predictive disasters.
- https://www.youtube.com/watch?v=pY_9TxAg95M



IoT in Government and Military

- Terror threat analysis through IoT device.
- Sensor bombs deployed on the battlefield to form sensor networks to monitor threats.
- Government asset tracking systems.
- Real-time military personal tracking and location services.
- Synthetic sensors to monitor hostile environments.
- Water level monitoring to measure dam and flood containment.
- https://www.youtube.com/watch?v=WKwjL2Y3ZGQ





Reference

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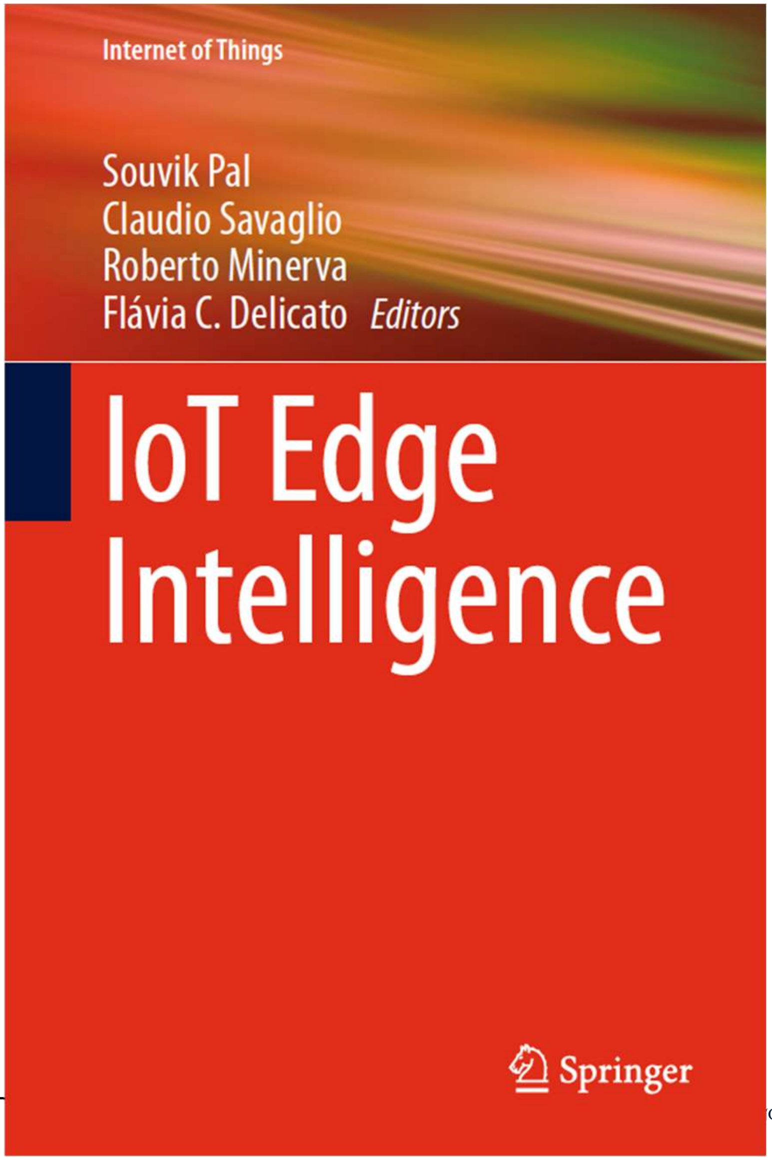

Digital Twin Technology

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Networking

Outline

- Digital Twin Introduction
- Digital Twin Technology
- Digital Twin Types
- Applications

What is a Digital Twin?

- A digital twin is a virtual representation of a physical object or system across its lifecycle, using real-time data to enable understanding, learning and reasoning. **IBM**

- A digital twin is a virtual representation of a physical product or process, used to understand and predict the physical counterpart's performance characteristics. **Siemens**

What is a Digital Twin?

- Digital twins are realistic digital representations of physical things. They unlock value by enabling improved insights that support better decisions, leading to better outcomes in the physical world. **Cambridge Centre for Digital Build Britain**

- A digital twin is a mirror image of a physical process that is articulated alongside the process in question, usually matching exactly the operation of the physical process which takes place in real-time. **Michael Batty, UCL**

What is a Digital Twin?

- The concept of Digital Twin was first used in Davin gall Inger's 1999 book called mirror worlds.

- Digital twin technology evolved from computer simulation and digital modeling, where the first use of the terminology "Digital Twin" was used in the "2010 Roadmap Report" from the National Aeronautics and Space Administration (NASA). Related terminology had appeared in earlier documents using slightly different names like "virtual twin" et al.

What is a Digital Twin?

- Actual systems can be modeled into a virtual digital form that has all the same essential properties such that testing, analysis, and modeling can be accurately conducted. Such systems are called Digital Twins.

- Digital Twins are an essential part of modern designing as accurate performance estimations and predictions can be made.

- Using advanced digital twins technology, testing, controlling, monitoring, integration, decomposition, maintenance, and modifying can be possible.

Why Require Digital Twin?

- It helps in designing and optimization part.
- It helps to predict the aging effect.
- It is capable of monitoring the performance of the physical object throughout the whole life cycle.
- A digital twin also integrate historical data from past machine usage to factor into its digital model.

The Digital Twin Model

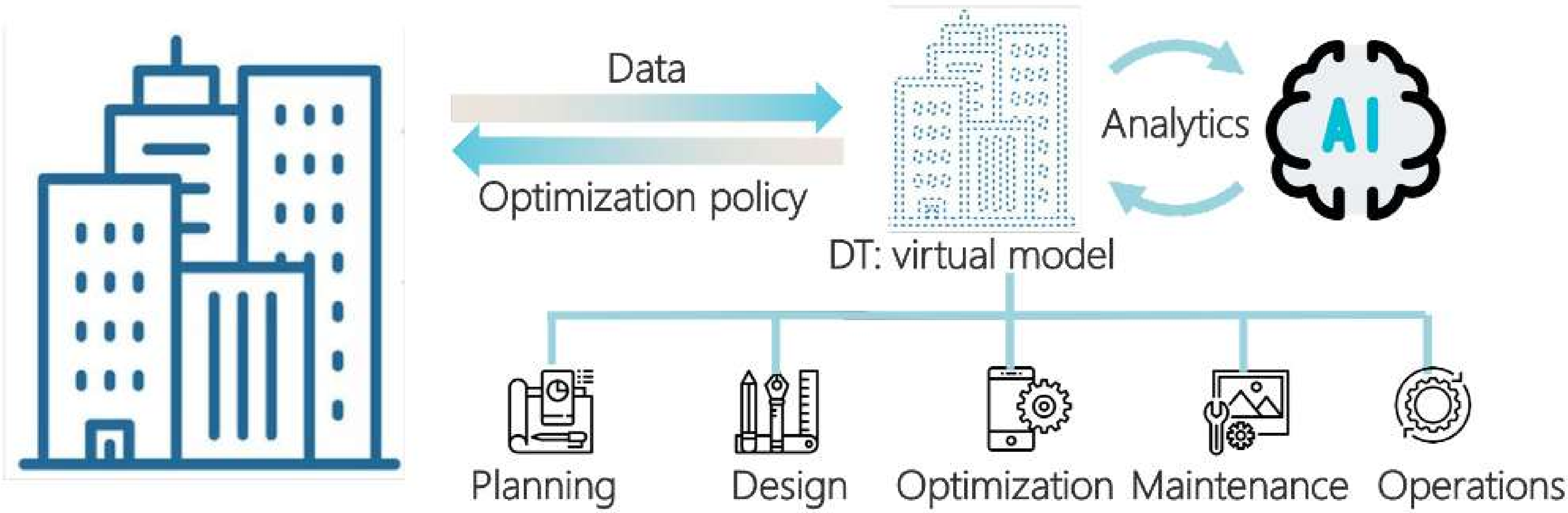

Fig. 1.3 Illustration of the digital twin model

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The Digital Twin Model

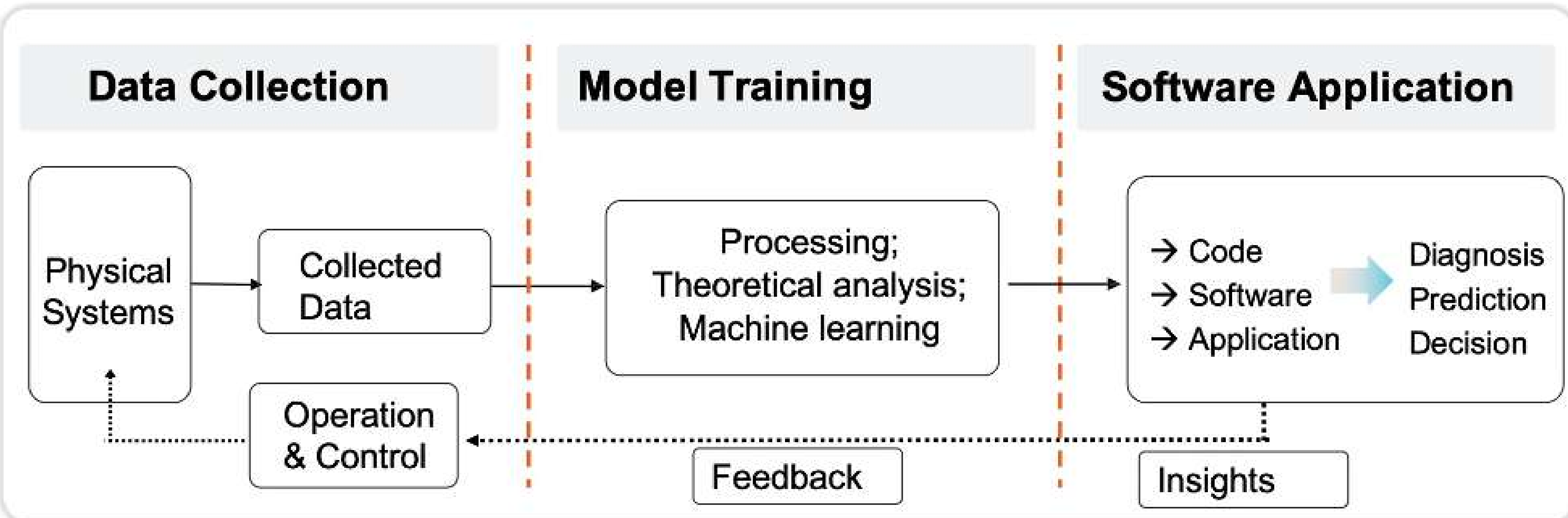

Fig. 1.4 Digital twin: data–model–software

The Digital Twin Model

- Data, the foundation
  - Data are the foundation of the entire digital twin model.
- The model, the core component
  - The theory models are constructed based on the principles from the physical systems.
  - The data-driven models are trained.
- Software, the essential carrier
  - Software is the carrier of digital twins and provides the interface between physical systems and digital space.

Standardized Guidelines for Digital Twin

- A digital twin will need to include a virtual model of all essential parts of the actual system such that an accurate behavior model can be created. Since actual physical systems commonly include modules and parts made from various companies, digital twins commonly need its parts to be also designed by multiple programmers and designers, which may be working for different companies/organizations. To make the virtual modules and parts that were designed from other companies/organizations work effectively together, digital twin modeling requires standardized guidelines.

Standardized Guidelines for Digital Twin

- Digital twin technologies need to be developed to fit these guidelines, so they can be integrated together in an efficient and reliable way. For this reason, the International Organization for Standardization (ISO) has created the **ISO 23247 standard digital twin framework** for manufacturing, which focuses on automation systems and integration.
- The ISO 23247 standard series are not the only standards for digital twin modeling. For example, the ISO 15926 series can be used for digital twin modeling for the oil and gas industry, and the ISO 16739 series can be used for digital twin modeling for building and constructions, et al.

Example

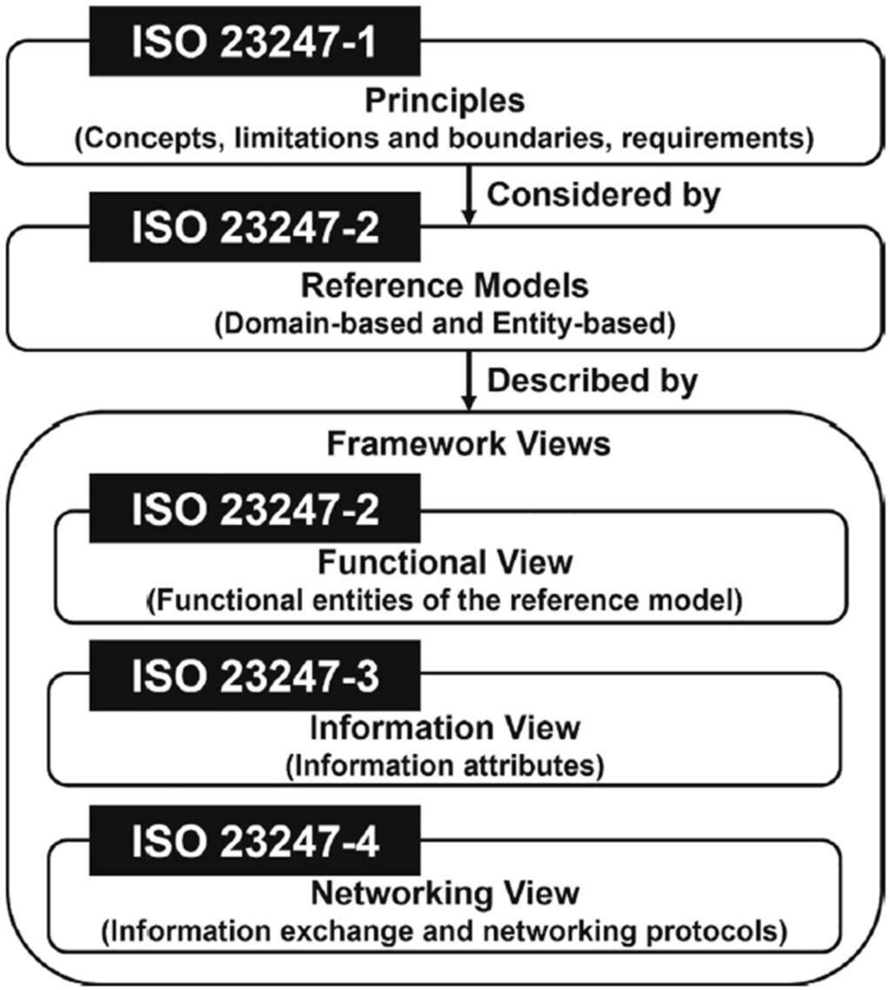

Fig. 2.26 Digital twin framework for manufacturing ISO 23247 series structure

Standardized Guidelines for Digital Twin

- The first part ISO 23247-1
  - “General principles and requirements for developing digital twins in manufacturing,” which covers the concept and requirements as well as the limitations and boundaries.
- The second part ISO 23247-2
  - “Reference architecture with functional views,” which include domain and entity references to be used for digital twin models.
- The third part ISO 23247-3
  - “List of basic information attributes for the observable manufacturing elements,” which provide details on how the information attributes of the digital twin’s information view can be modeled.
- The fourth part ISO 23247-4
  - “Technical requirements for information exchange between entities within the reference architecture,” which describes the information exchange and networking protocols that need to be designed to support the digital twin’s networking view for automation systems and integration manufacturing.

Digital Twin Framework

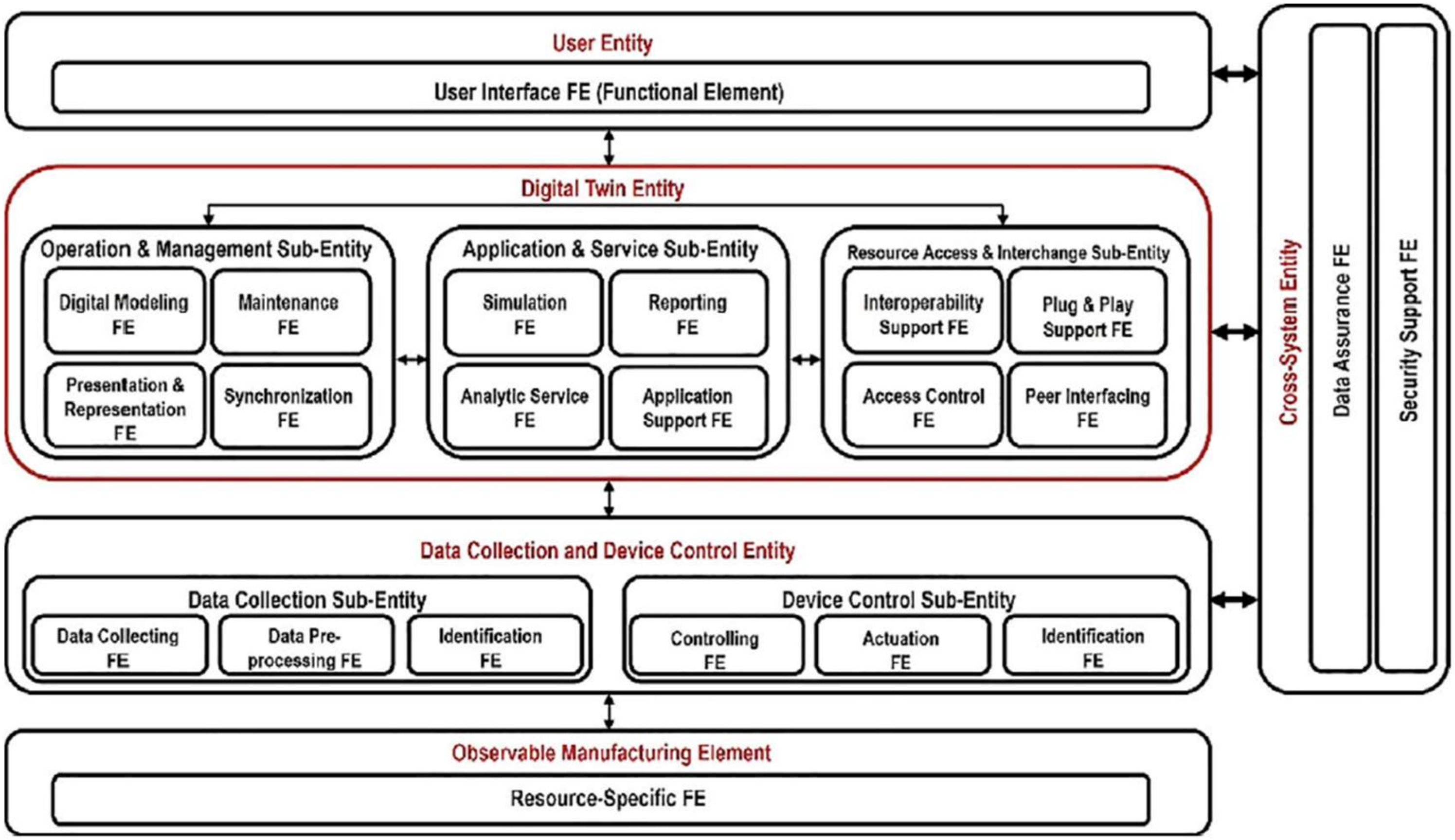

Fig. 2.27 Structure of the ISO 23247 digital twin framework for manufacturing

Digital Twin Framework

- The **Digital Twin Entity** is the center block, which is the center of the modeling process.

- The digital twin entity includes three main units, which are the **Operation and Management Sub-Entity**, **Application and Service Sub-Entity**, and **the Resource Access and Interchange Sub-Entity**.
  - The three sub-entities and their functional elements (FEs) must be interconnected so information exchange and functional collaboration is possible at all sub-layer and sub-domains in the digital twin model

Digital Twin Framework

- Three sub-entities of the digital twin entity:
  - **Operation and Management Sub-Entity** includes functional elements for digital modeling, maintenance, presentation and representation, and synchronization.
  - **Application and Service Sub-Entity** includes the functional elements for simulation, reporting, analytic services, and application support.
  - **The Resource Access and Interchange Sub-Entity** includes the functional elements for interoperability support, plug-and-play support, access control, and peer interfacing.

Digital Twin Framework

- Digital Twin Entity interfaces with several entities:
  - **User Entity** through the user interface functional element.
  - **Data Collection and Device Control Entity**, which is consists of two sub-entities:
    - The Data Collection Sub-Entity includes the functional elements for data collecting, data pre-processing, and identification.
    - The Device Control Sub-Entity includes the functional elements for controlling, actuation, and identification.
- The **Data Collection and Device Control Entity** interfaces with the **Observable Manufacturing Element**, which include the resource-specific functional elements.

Digital Twin Framework

- The User Entity, Digital Twin Entity, and the Data Collection and Device Control Entity interfaces with the Cross-System Entity to enable efficient, reliable, and secure collaboration and coordination.

- The **Cross-System Entity** includes the functional elements for data assurance and security support.
  - The **data assurance functional element** includes support for in-time delivery of important information as well as data source reliability support, which include data access authorization as well as data authentication.
  - The **security support functional element** includes various cyber-attack protection functions, anomaly detection, as well as counter defense measures to protect the digital twin network and operational entities.

Features of Digital Twin

- Connectivity
  - A digital twin is based on connectivity. It enables connection between the physical element and its digital counterparts. The sensors create the connectivity of physical products that obtain, integrate and communicate data using various integration technologies.

- Homogenization
  - Homogenization has two component steps:
    - Spatially aligning data
    - Ensuring the quality of the data is compatible
  - Digital twins are both the consequence and enabler of homogenization of data. It allows the decoupling of information from its physical form.

Features of Digital Twin

- Digital traces
  - Digital twin technologies leave digital traces. The trails are helpful to diagnose the source of the problem that occurred in case of machine malfunctions.
- Modularity
  - Modularity is referred to the design and customization of the products and production modules. The addition of modularity to functional models helps manufacturers gain the ability to tweak machines and models.
- Reprogrammable and Smart
  - Digital twins automatically enable re-programmability through sensors, artificial intelligence techniques and predictive analysis

Advantages and Disadvantages

- Advantages
  - Provides a representation of operational data.
  - Analysis of historical trends and potential causes for performance deterioration.
  - Support for decision making and optimisation of day-to-day operations.
  - Identify abnormal behaviour and reduce downtime.
  - Modelling future behavior based on analysis of operational parameters.
  - Predictive maintenance to minimise operational disruptions.
  - Agile decision making in response to changes and interventions.
  - Support for long-term strategic decisions to enhance efficiency, safety, reliability.
  - Identifies and recommends new interventions and actions.
  - Reduced dependence on human operators – enhanced safety.

Advantages and Disadvantages

- Disadvantages
  - The success of technology is dependent on internet connectivity.
  - The security is at stake.
  - Digital twin will be required across entire supply chains.
  - The challenges involved here include globalization and new manufacturing techniques.

Digital Twin Types

- One way to classify the types of digital twins is：
  - Digital Twin Prototype (DTP)
  - Digital Twin Instance (DTI)
  - Digital Twin Aggregate (DTA)

Digital Twin Types - DTP

- **Digital Twin Prototype (DTP)** is used to help design and test a prototype of a physical system or service before it is released for official/commercial use by customers/users.
- DTPs need to include design features and behavioral characteristics that are sufficiently accurate such that the testing and analysis results of the DTP are reliable. DTPs enable corrections to be made before the product or service is released.
- A great benefit of a DTP is to be able to see how the physical system or service would perform and appear to its customers/users before it is officially made public for sale or service.

Digital Twin Types - DTI

- **Digital Twin Instance (DTI)** is used to check a product or service status once the product or services has been manufactured, released to the public, and/or in operation.
- A DTI will be used by the manufacturing company to check problems in a product that has been reported by its users, or to predict a failure due to upcoming events in which the product or service may be used. A manufacturer may not be able to predict all environments in which its products or services could be used in. So, as new situations arise, the manufacture or retailer would need a DTI to determine the level of performance its products or service would be able to provide in the new environment.

Digital Twin Types - DTI

- In addition, even if the environment has not changed, as the product or service ages, component failures or module deterioration can result in different characteristics and performance levels, in which a DTI would be able to significantly help to inform what parts need to be replaced or upgraded in advance to help prevent a failure. As such failures of a system or service can result in critical damage to property as well as result in human casualties, the value of an accurate DTI is significant.
- This is why DTIs are used by automobile and airplane companies, as they help to predict failures, which may lead to recalls for parts replacement so accidents can be prevented.

Digital Twin Types - DTA

- **Digital Twin Aggregate (DTA)** is needed when it is not be possible to create a DTI for a large system or a network of systems. A DTA is an aggregation of multiple DTIs in which the interconnected behavioral characteristics are integrated into the DTA such that the comprehensive virtual functionalities are accurately expressed.
- Evidently, a DTA takes much more effort to design and build than a DTI, due to the complex interconnected effects between the DTIs. A complex system prototype would also need a DTA to be created from the combination of its DTP and DTI modules, in which a network that characterizes the integrated interconnection of the DTPs and DTIs must be included.

Digital Twin Types - DTA

- Although a DTA is more difficult and expensive to design and build, a properly designed DTA would be able to help fix malfunctions, prevent large scale accidents, and predict methods to maximize the performance. This would result in prevention of large-scale system failures and security breaches, as well as help increase the performance of the networked systems in which savings and profits could be significant. In many cases, investments into accurate DTAs are worthwhile as causalities and/or major property loss can be prevented.

Digital Twin Maturity Levels

- Level 1 - Basic Reporting and Analysis
  - Dynamic low fidelity model
  - Linked to the real-world system
  - Model updated at regular intervals but not necessarily in real-time
  - No capability for autonomous decision making
  - Basic learning, data assimilation capability
  - Basic analytics and visualisation functionality of the current state

Digital Twin Maturity Levels

- Level 2 - Advanced Reporting, Analysis and Forecasting
  - Real-time, dynamic high fidelity models developed from first principles
  - Model updated in real-time
  - Standard pre-defined control/decision-making capability
  - Advanced data assimilation capability to ensure convergence and accuracy
  - Advanced predictive analytics

Digital Twin Maturity Levels

- Level 3 - Support for Strategic Decision Making
  - Real-time, dynamic, multi-fidelity hybrid models of cyber-physical systems
  - Real-time operation
  - Advanced, adaptive control/decision-making. Can recommend interventions
  - within a particular domain.
  - Advanced spatio-temporal analytics and visualisation
  - Advanced learning and data assimilation from multiple data sets

Digital Twin Maturity Levels

- Level 4 - Autonomous Decision Making
  - Dynamic multi-scale, multi-domain, multi-system models - federated digital twins
  - Real time operation
  - High-level of autonomy – can perform most human operator functions, make
  - interventions and take new courses of action autonomously
  - State-of-the-art analytics and visualisation
  - State-of-the-art AI/machine learning capability

Applications

- Manufacturing
  - In the manufacturing industry, digital twins are used for facilitating product development, design customization, shop floor performance improvement and predictive maintenance.

- Product Development
  - Engineers gain benefits from digital twins as it helps to test the feasibility of upcoming products before launching.

Applications

- Shop floor performance improvement
  - Digital twins are helpful in monitoring and analyzing the end products. It helps engineers to spot the defective and low-performing products in the lot.

- Predictive maintenance
  - Digital twins help manufactures predict potential downtimes of machines to improve their overall productivity by minimizing non-value-adding maintenance activities.

Applications

- Retail
  - In the retail sector, digital twins are used for modelling and augment customer experience at prominent shopping centers and individual stores.

- Automotive
  - Digital twins are highly used for creating virtual models of connected vehicles. Automotive companies simulate and analyze the production phase to identify the potential problems during production or when the car hits the roads.

Applications

- Healthcare
  - Digital twins virtualize healthcare services and help healthcare providers to optimize patent care, cost and performance. It aims to improve the operational efficiency of healthcare processes and enhance personalized care.

- Smart cities
  - The digital twin can help cities to become more economically, socially and environmentally sustainable. Virtual models can guide planning decision and offer solutions to the many complex challenges faced by modern cities.

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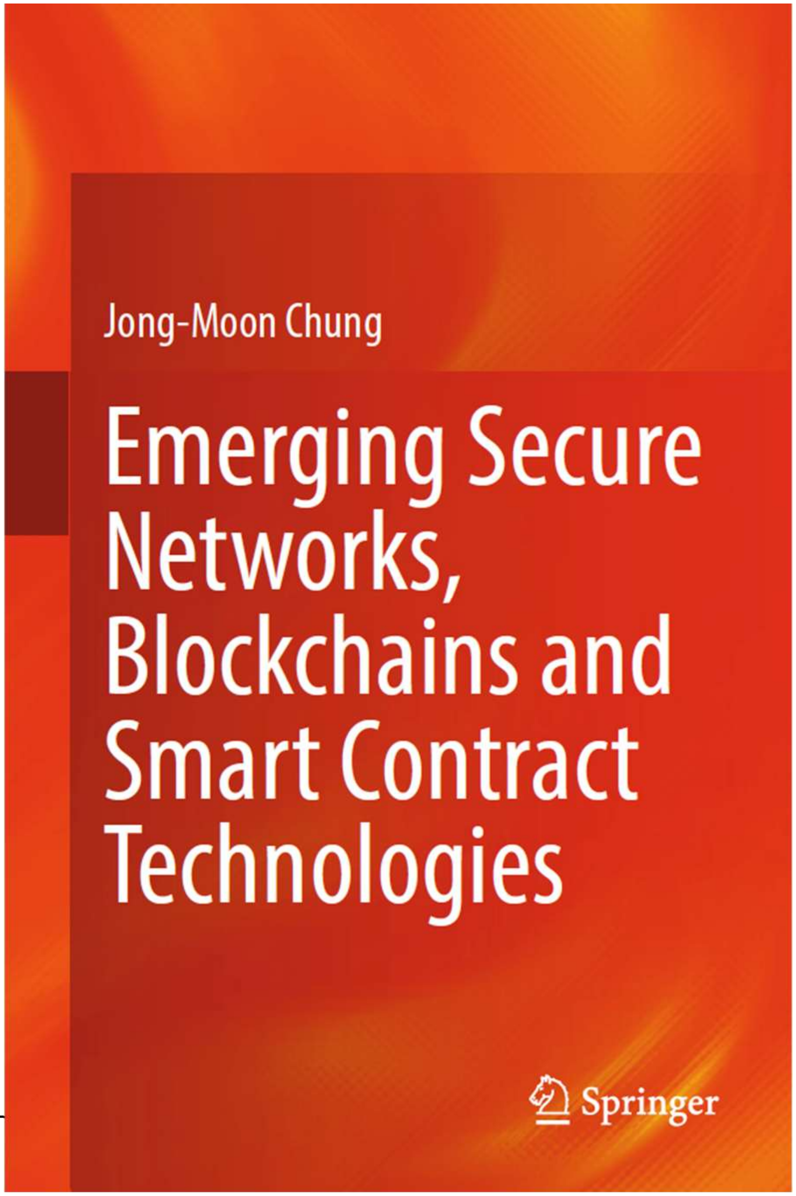

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